

Vikram Singh

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PROFESSIONAL SUMMARY

Mechanical Design Engineer with 5 years of experience in product design, CAD modeling, and manufacturing process optimization within the automotive components industry. Proven ability to reduce production costs while maintaining strict quality standards.

SKILLS

SolidWorks	AutoCAD	CATIA V5
ANSYS (FEA)	GD&T	DFMEA / PFMEA
Six Sigma (Green Belt)	Lean Manufacturing	CNC Machining
Tolerance Stack-up Analysis	Material Selection	Product Lifecycle Management (PLM)
Root Cause Analysis	MS Project	ISO 9001 Quality Standards

WORK EXPERIENCE

Senior Mechanical Design Engineer — Apex Auto Components Ltd

Coimbatore, India | Feb 2022 – Present

- Led design of a lightweight suspension bracket that reduced component weight by 15% without compromising strength.
- Conducted FEA simulations in ANSYS to validate designs, reducing physical prototype iterations by 40%.
- Implemented Lean manufacturing changes on the assembly line that improved throughput by 20%.
- Coordinated with cross-functional teams to launch 4 new product lines within budget and schedule.

Design Engineer — Tristar Precision Engineering

Chennai, India | Jun 2019 – Jan 2022

- Created detailed 3D models and production drawings in SolidWorks for automotive sub-assemblies.
- Performed tolerance stack-up analysis to resolve recurring fitment issues, cutting rework by 25%.
- Supported DFMEA reviews that identified critical failure modes prior to production release.

EDUCATION

B.E. in Mechanical Engineering

PSG College of Technology, Coimbatore | 2015 – 2019

CERTIFICATIONS

- Six Sigma Green Belt Certification (2021)
- Certified SolidWorks Professional (CSWP), 2020